BO YANG

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EDUCATION

University of California, Los Angeles, Ph.D. in Civil Engineering (Transportation) — Sep. 2025 – Present
National University of Singapore, Master of Computing in Computer Science (CAP: 4.15/5) — Jan. 2022 – Jul. 2023
Nanjing University, B.Eng. in Industrial Engineering (GPA: 4.36/5) — Sep. 2017 – Jul. 2021

RESEARCH FOCUS

Generative modeling of multi-level human mobility, from macroscopic travel behavior to microscopic trajectories.

RESEARCH EXPERIENCE

UCLA Mobility Lab, Graduate Student Researcher (Advisor: Prof. Jiaqi Ma) — Sep. 2025 – Present

Diffusion models for whole-day activity generation — ongoing

- Built a conditional diffusion model with a Transformer denoiser that jointly generates location, activity type, and timing for a full day (5,697 zones × 15 activities × 96 slots), trained on 8.1M person-days from 618K GPS users in Los Angeles.
- Outperformed 8 generative baselines (incl. MoveSim, TimeGeo, ActVAE, TrajGPT, MIRAGE) on 5 of 6 held-out realism metrics, with 45% and 44% lower divergence in activity duration and activity-chain structure than the strongest baseline.
- Developed maximum-entropy guidance that adapts the trained model to a new year using only published aggregate statistics, matching fine-tuning on individual-level data and cutting distance error in unseen regions from 14.1 to 1.1 km.

Trip purpose inference from GPS trajectories — [1], co-first author

- Designed a weakly supervised, uncertainty-aware pipeline that infers trip purposes for 81M GPS staypoints without ground-truth labels, combining LLM-enriched POI semantic zones, Bayesian activity scoring, and multi-phase Pareto calibration to travel-survey statistics; cut divergence in activity type, start time, and duration by 23%, 48%, and 12%.

Household-coordinated population mobility generation (PopTraj) — [3], co-author

- Led census-constrained population synthesis and home/work/school anchor assignment for a hierarchical pipeline that distills an LLM teacher into a hybrid CVAE, generating 395K household-coordinated person-weeks in Tokyo with 39% lower divergence than the best of 9 baselines.

MIT-NEU-UF Research Program, Advisors: Prof. Jinhua Zhao & Dr. Yuebing Liang — May 2024 – Jun. 2025

Deep reinforcement learning for transit-oriented development planning — [4], first author

- Formulated station-area land-use planning as sequential decision-making on a parcel–road–station graph, with a combinatorial action space (land use × floor-area ratio) and a multi-objective reward encoding TOD indicators and reconstruction cost.
- Improved spatial efficiency by 70% and 54% over rule-based and genetic-algorithm baselines in a Shanghai case study.

NUS Master's Capstone, vision-language pipeline for dashcam driving-style profiling — Feb. 2023 – May 2023
NJU Undergraduate Thesis, deep RL with a Transformer policy for TSP local search — Jan. 2021 – Jun. 2021

PUBLICATIONS (* EQUAL CONTRIBUTION)

[1] **Yang, B.***, Ma, H.*, Liu, Y., Zhang, Z., Stanford, C., Sun, M., & Ma, J. Uncertainty-Aware Trip Purpose Inference from GPS Trajectories via POI Semantic Zones and Pareto Calibration. *IEEE ITSC*, 2026. [arXiv]

[2] Liu, Y.*, Sang, Y.*, Liao, X., Sun, M., **Yang, B.**, Zhang, Z., Stanford, C., Ma, H., & Ma, J. GPS-Enhanced Tourist Mobility Modeling with Seasonal Spatial Priors and LLM-Based Activity Chain Generation. *IEEE ITSC*, 2026. [arXiv]

[3] Ma, H., Liu, Y., **Yang, B.**, Zhang, Y., Stanford, C., Fox, E., Liao, X., & Ma, J. PopTraj: Hierarchical Multi-Source Generation of Household-Coordinated Weekly Activity–Location Trajectories with Urban Population Synthesis. *Information Fusion*, under review (2nd round). [SSRN]

[4] **Yang, B.**, Liang, Y., Zhao, Z., Ding, F., Zheng, Y., Wang, S., & Zhao, J. Performing Transit-Oriented Development Planning Using Deep Reinforcement Learning. *Land Use Policy*, under review (2nd round).

ADDITIONAL EXPERIENCE

Temasek Labs @ NUS, Graduate Researcher: ANN vector search for radar-signal retrieval — Jun. 2022 – Sep. 2022
NUS Smart Medical Tech Lab, Research Intern: vision-based vital-signs detection — Sep. 2021 – Dec. 2021

SERVICE & SKILLS

Service: Editorial Coordinator, IEEE Open Journal of ITS (OJ-ITS); Student Committee Member, IEEE ITSS Technical Committee on Trustworthy AI
ML: Diffusion models, VAEs/CVAEs, Transformers, LLM prompting and distillation, deep RL, constrained/guided generation
Programming: Python, PyTorch, SQL; also TensorFlow, MATLAB, R, Java, C